COUNTING FROM THE FIRST TICK

Life, the Universe, and God

A Software Engineer's Instrument for the Immeasurable

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2026 · built against the ucal source tree

Contents

Before we begin

Three things are true about this project, and the third only follows if you accept the first two. So they go here, on the first page, before anything has a chance to soften them.

The artifact is real. `ucal` is working Rust — six crates, a library and a command line, published and installable. Time is an unsigned integer count of Planck-time units since a stipulated datum. There is no floating-point value anywhere in the workspace, not in a signature, a field, an intermediate, or the rendering path. Every constant in it is reproducible by two independent derivations that agree bit for bit. At the commit this book is pinned to, 381 tests pass on both integer backends. You can check every sentence in that paragraph; the last page of this book tells you how.

Its practical utility is questionable. No task you have today needs a Planck-tick count since a stipulated datum. It does not compete with `chrono`, `time`, or `hifitime`, and it does not try to. Nothing on its ladder of units is near a second or an hour — a second is 21.4 beats, an hour is 24.6 arcs — so it is not merely unnecessary for ordinary work but actively awkward for it. There is exactly one qualification to this, it concerns timekeeping across more than one planet, it takes up a single page in Part V, and it is not a rescue.

Therefore it is research of another kind. Art, philosophy, theology — conducted in the medium of a working program. That is not a consolation prize awarded after usefulness failed. It is what the thing is.

The rigor is the medium

Here is the part that matters, and it is easy to misread as modesty.

That the code compiles, that the constants reproduce, that the tests pass — these are not preliminaries to the argument. They **are** the argument, in the same way that the metre of a poem is not preliminary to the poem.

An essay can assert that a distinction ought to be respected. It can argue at length, and persuade you, and then it is over, and the discipline it recommended survives exactly as long as the next author's attention does. A type system can do something an essay cannot: it can make violating the distinction **fail to build**. Not discouraged. Not deprecated with a warning. Refused, by a machine, to a person who disagrees.

That difference is the thesis of this book, and the reason it is a book about a program rather than a program with a book attached.

What this is not

Some of these will be obvious. They are listed because a book that surveys nine philosophical and religious traditions alongside a Rust crate will be misread in predictable directions, and it is cheaper to close those doors now than to keep apologising later.

Not an apologetic No tradition is argued true here. The code is evidence for none of them. Where a tradition and the instrument agree, that is reported as a convergence and never as a vindication of either.

Not a proof of usefulness The second fact above stands unretracted.

Not a claim of discovery When a tenth-century argument turns out to match a twenty-first-century type signature, the older text is not thereby shown to have anticipated anything. Convergence is not anticipation.

Not numerology No constant, base, or magnitude in this system acquires meaning by resembling a number in a tradition. Base 5 was chosen because $5^5 = 3125$ gives a tier of exactly five digits. That is the whole reason. The book says so plainly and then reports, as a curiosity and not as evidence, that the first place the research went looking for a five, it found one waiting.

Not a tutorial If you want to use the crate, its documentation is better at that than this book will be.

Not a defence of the datum Tick zero is stipulated. That it is stipulated is the point, and Part IV is four chapters about why.

Two paths through the book

The book serves two readers who are rarely the same person, and it does not ask either of them to pretend to be the other.

The **engineer's path** is Parts I, II, III, and V. It is a complete technical article about an unusual piece of software: what it computes, what its type system refuses, where the design lost arguments with the compiler, and how far the approach generalises beyond Earth. You can stop at the end of Part V and have received the whole of that.

The **reader's path** is Parts I, IV, VI, VII, and VIII. It is a book about what it means to build a measuring instrument that points at something it declares itself unable to measure, and about nine traditions that have thought carefully about origins, number, and time — read here as readers of the artifact rather than as authorities over it.

Part I belongs to both. Nothing later in the book may redefine what Part I establishes.

> **A NOTE ON MARKING**
>
> This book marks its own claims. Where a passage is **interpretation** — the author reading a meaning into the artifact — it sits in a ruled block that says so. Where a coincidence is genuinely striking but proves nothing, it is labelled **resonance**, and it may never appear as a premise for anything.
>
> That apparatus is not decoration. It is the book submitting to the discipline it documents: the software marks where it stops computing, and the book marks where it stops asserting fact. If you delete every marked block, every technical claim in Parts I through III and V must still stand. Making sure that is true is a scheduled step in producing this book, not a hope about it.

A word about being wrong

The most valuable document in this project's history is a correction.

Two revisions of the specification asserted that the continued-fraction machinery reproduces the Julian and Gregorian calendar rules as convergents. The Julian rule does appear — it is the first convergent. The Gregorian rule does not appear at any depth, and a rule twelve times simpler is more accurate than it. The machinery contradicted its author, in writing, on a claim he had published twice.

That correction is in this book, dated, including the revisions that were wrong. It is in here because a system built to make its author's conclusions provable would never have produced it, and because the charge that such systems are **always** built that way is nine hundred years old and deserves an answer made of evidence rather than assurance.

If you find yourself reading a passage that seems to be presenting that correction as foresight rather than as an error published — a demonstration of rigour rather than an argument the author lost — then the book has failed at the thing it cares most about, and you should say so.

PART I

Foundations

Definitional. Nothing later in the book may redefine these, and a reader who stops here must not leave with a false impression.

CHAPTER 1

1

Intent

This began as an irritation, and it is worth being honest that it began as an engineer's irritation rather than a philosopher's question.

Read almost any account of the early universe and you will find a sentence like this one: **recombination occurred about 380,000 years after the Big Bang.** It is a good sentence. It is doing real work. And if you stop and ask what its units are, something uncomfortable happens.

A year is the time Earth takes to go around the Sun. Not approximately — definitionally. The Julian year used in astronomy is exactly 365.25 days of exactly 86,400 seconds, and those numbers are what they are because of how fast one particular rock spins and how long it takes to circle one particular star. So **380,000 years after the Big Bang** means 380,000 × 365.25 × 86,400 seconds, which means it is a quantity expressed in units defined by the rotation and orbit of a planet that would not exist for another nine billion years.

The number is correct. It is not wrong in any way that matters to a cosmologist. But it carries a passenger, and the passenger is Earth.

The irritation, stated precisely

The problem is not that the units are arbitrary. All units are arbitrary. The metre is a bar in a vault, then a wavelength, then a fraction of a light-second; none of that makes it a bad metre.

The problem is **provenance leaking into arithmetic**. When you compute with Earth years, Earth's orbital period is inside every intermediate value. Usually that costs nothing. Occasionally it costs you a rounding you did not intend, because 365.25 days is not a whole number of anything and the conversions do not close. And structurally — which is the part that would not let go — it means the description of an event 13.8 billion years old is phrased in terms of a body that had no bearing on it.

> **WHAT THIS IS NOT AN ARGUMENT FOR**
>
> None of this is a criticism of how cosmology is done. Astronomers use Julian years because they are stable, agreed, and unambiguous, which is exactly what a unit should be. The complaint here is narrow and it is aesthetic before it is technical: a quantity ought to be expressible in units that do not smuggle in a planet.

What would have to be true instead

Suppose you wanted a time system with no Earth content in its arithmetic. What would it need?

It would need a **unit** that is not defined by any body's motion — something built out of constants of nature rather than out of a rotation.

It would need an **origin** that is not a calendar event, since every calendar epoch is someone's civil history.

It would need a **notation** in which writing fewer digits means saying less precisely, rather than meaning zero — because at these magnitudes you will constantly be quoting numbers you do not know to full precision, and a system that silently pads them with zeros is lying on your behalf.

And it would need to be honest about the one thing it cannot do: it cannot tell you what time it is on Earth without, at some point, being told about Earth. That contact has to happen somewhere. The question is whether it happens **in the arithmetic** or at a declared boundary you can point to.

That last question turned out to be the whole design.

What was built

`ucal` is the answer to those four requirements, and Part II is the technical account of it. In one paragraph: absolute time is an unsigned integer count of Planck-time units since a stipulated datum; those ticks are grouped in a ladder of powers of five, so a timestamp is the tick count written in base 5 and truncating it **is** rounding it; Earth enters through exactly one declared constant and leaves at the formatting boundary; and no floating-point value appears anywhere in the workspace.

It works. It is also, as the preface said and will keep saying, of questionable use.

Where the trouble started

The requirement that has no clean answer is the second one: the origin.

If time is a count since some zero, the zero has to be somewhere. And the obvious place to put it — the beginning — is not available, for reasons that took four chapters of this book to state properly and that Part IV is entirely about.

The short version, so you can carry it through Parts II and III: **the age of the universe is a measurement, and measurements have error bars, and an error bar cannot be the origin of an exact integer count.** The published figure is 13.787 billion years give or take 0.020 billion. That uncertainty, converted to ticks, is a number with fifty-eight digits. If the datum inherited it, every timestamp in the system would inherit it too, and the exactness the whole design exists to protect would be theatre.

So the datum is **stipulated**. Declared, not discovered. Tick zero is a reference point that the specification says, in as many words, is not a measurement and not an observed event.

And then the interesting problem: how do you say that in a way that a future maintainer — or the author, three years later, in a hurry — cannot quietly forget?

> INTERPRETATION
>
> The answer this project arrived at is the reason it turned into a book. The physical claim about the origin is declared, cited, and given an exact magnitude. And then it is made **impossible to compute with**: a type with no arithmetic operations at all, guarded by a test that fails to compile if the type ever reaches an operand position.
>
> You cannot forget a discipline that the compiler enforces. That is a small technical fact with a large philosophical shadow, and the shadow is what Parts IV, VII, and VIII are about.

What this book will and will not claim

Before Part II starts and the prose becomes technical for a hundred pages, here is the calibration.

It **will** claim that a distinction enforced in a type system is a philosophical contribution and not merely an implementation detail. It will claim that the approach generalises to any celestial body, and it will spend a chapter on exactly where it fails. It will claim that the instrument does philosophical work no prose argument can do, and it will demonstrate that rather than assert it.

It will **not** claim that time began, that any tradition is correct, that the base is meaningful, or that anyone should adopt this calendar.

WHAT THIS CHAPTER ESTABLISHED

- Cosmic durations are conventionally expressed in units defined by one planet's motion — correct, useful, and carrying a passenger.
- A time system with no Earth content in its arithmetic needs a physical unit, a non-calendrical origin, a notation where truncation means imprecision, and one declared point of contact with Earth.
- The origin cannot be measured, because a measurement has error bars and an exact integer count cannot inherit one.
- So it is stipulated — and the project's contribution is making that stipulation mechanically impossible to forget.

CHAPTER 2

2

The tick

Everything this system computes is an unsigned integer count of ticks. Nothing else is primitive — not the second, not the beat, not the day. If you understand why the tick was chosen and, more importantly, what choosing it does **not** claim, you have the foundation of the whole instrument.

TERM **Tick**

The Planck time, $t_P = \sqrt{\hbar G/c^5} \approx 5.391 \times 10^{-44}$ seconds. In this system it is the atomic unit of duration: the smallest quantity representable, and the unit in which every other quantity is counted.

Why this unit

The Planck time is composed from three constants: the gravitational constant G, the reduced Planck constant $\hbar$, and the speed of light c. Those three bound gravitation, quantum action, and the propagation of causality respectively — and there is exactly one combination of them with the dimension of time.

That is the appeal, and it is worth stating carefully because it is easy to overstate. The tick is not defined by any body's motion. It does not know about Earth, or about the Solar System, or about matter being organised into planets at all. It is the interval you get when you ask the three limiting constants of physics what a duration would be, and they answer.

For a system whose entire purpose is to have no Earth content in its arithmetic, that is the right floor.

What the tick is not

Here is where a great deal of writing about the Planck time goes wrong, and where this project declines to follow.

The tick is not a quantum of time. It is the resolution floor of an instrument, not a structure discovered in the world. Nothing in this project asserts that time comes in discrete lumps, that there is a smallest possible interval, or that asking about a shorter duration is physically meaningless. The system cannot represent a shorter duration; that is a fact about the system.

The distinction matters more than it looks, because asserting otherwise would silently take a side in an argument that has been running for twenty-four centuries.

> Aristotle holds in **Physics** VI that time is a continuum: divisible without limit, and the instant is a **limit** of an interval rather than a part of it, in the way that a

> point is a limit of a line and not a very short line. On this view a smallest duration is not merely unobserved but incoherent.
>
> Against that, temporal atomism has a long and respectable history. Epicurus held that there are **minima** of time as there are minima of magnitude. The Kalām tradition developed a thoroughgoing atomism in which time is composed of indivisible moments and accidents do not endure from one to the next.

The instrument is discrete. If discreteness were asserted as a claim about the world, this project would have joined Epicurus and the Kalām against Aristotle — and it would have done so not on the strength of an argument but as a side effect of choosing an integer type.

That is not a respectable way to take a metaphysical position. So the position is declined, explicitly, and the declining is written down: the tick is where the instrument's resolution stops, and the instrument makes no claim about where the world's does.

> A RULE YOU WILL SEE AGAIN
>
> This pattern — **the implementation forces a choice; name the choice; refuse the claim that would come free with it** — recurs throughout the design. It is the same move that Part IV makes about the datum, at much greater length and with much higher stakes.

The concession: the tick's length

There is one place where the tick is not free of Earth, and it should be admitted here rather than discovered later.

The Planck time's **numerical value** is known only as well as G is measured, and G is the worst-measured of the fundamental constants by a wide margin. Worse, expressing it at all requires a unit of time to express it **in** — and that unit is the SI second, which is defined by a caesium transition counted by instruments on Earth.

So the tick's length is fixed by convention against the second. The system declares a value, records where the value came from, and uses that declared value everywhere. It does not re-derive it, and it does not pretend the value is more certain than the measurement behind it.

INTERPRETATION

What this concedes is **metrology** and nothing else. The arithmetic contains no Earth content: ticks are counted, not converted. What is Earth-flavoured is the statement of how long a tick is in seconds — which is a sentence about translating between two systems, not a fact used inside either.

Whether that is a satisfying answer is a fair question, and Part III returns to it. It is at least an honest one, and its honesty is structural: the conversion lives at a declared boundary you can point to rather than dissolved into the operations.

Counting, not measuring

One more consequence, which will matter in Part IV.

Because time here is a **count**, the system's exactness is the exactness of integer arithmetic. Tick 8,070,204,002,895,596,515,944,343,085,635,637,180,530,466,139,316,558,837,890,625 is a specific tick, and adding one to it gives the next one, and that is all perfectly exact — in the same way that the number of steps you have walked is exact whether or not you know it.

What is **not** exact, and cannot be made exact by any amount of careful arithmetic, is the correspondence between tick zero and any physical event. That is a separate question, it has a separate answer, and keeping those two things apart is the discipline the rest of this book is about.

WHAT THIS CHAPTER ESTABLISHED

- The tick is the Planck time, built from G, $\hbar$, and c — the one combination of the three limiting constants with the dimension of time.
- It is the resolution floor of an instrument, **not** a quantum of time. Asserting otherwise would take a side in a live argument as a side effect of choosing an integer type.
- Its length in seconds is fixed by convention against SI. This concedes metrology and nothing else, at a boundary you can point to.
- Time is counted, not measured. The count is exact; what tick zero corresponds to is a different question entirely.

CHAPTER 3

3

The zero of time

A count needs somewhere to start counting from. This chapter says where, and says the one thing about it you need in order to read Parts II and III without being misled. The full argument is Part IV, four chapters of it, and the deferral is deliberate — but a reader who stops at the end of Part I must not leave with a false impression, so the essential point is made here and made plainly.

TERM **Datum**

Tick zero. A **stipulated** reference point, conventionally identified with the Friedmann–Lemaître–Robertson–Walker $t \to 0$ limit. It is not a measurement and not an observed event.

The domain is unsigned

Absolute time in this system is an unsigned integer. There is no tick −1.

This is not a storage optimisation. It is a statement about what the system is willing to represent: nothing precedes the datum, because the datum is where the count begins, and a count does not go backwards past its own start.

Ask for an instant before tick zero and you do not get a negative number. You get an error — `UCAL-E0020` — and the operation fails.

INTERPRETATION

There is a difference between answering a question with a negative number and refusing the question, and the system takes the second option deliberately.

A negative tick count would be an **answer**: it would say that the time before the datum exists, is measurable, and happens to lie on the other side of a chosen origin — the way 1 BC lies on the other side of the Gregorian epoch. The refusal says something else: that the question, as posed to this instrument, is malformed.

Augustine gives the classic form of the move in **Confessions** XI, answering what God was doing before creating heaven and earth. His answer is not a description of that interval but a rejection of the question's shape, because time is among the things created and there is no "before" of the sort the question wants. Whether or not you find the theology congenial, the logical structure is exactly `UCAL-E0020`: an error, not a negative number.

Why it is stipulated and not measured

Here is the one-line version. Part IV gives it four chapters and three further reasons; this is the one you can carry.

Exactness cannot come from measurement.

The age of the universe is a measured quantity. The current best figure is 13.787 billion years, plus or minus 0.020 billion. That is an excellent measurement. It is also, like every measurement, an interval rather than a point.

Convert that uncertainty into ticks and it becomes:

ucal datum — the claim's magnitude

```
big_bang_claim:
  half_width_ticks
1170697614114106987200000000000000000000000000000000000000
  half_width_drifts  141.53
```

Fifty-eight digits. About 0.145% of the whole span from the datum to now.

Now suppose the datum were defined **as** the measured age. Every timestamp in the system would inherit that uncertainty, because every timestamp is an offset from the datum. The exact integer arithmetic — the thing this entire design exists to protect — would be computing precise-looking answers on top of a foundation that wobbles by a number with fifty-eight digits in it.

You cannot get an exact origin from an inexact measurement. So the origin is not taken from the measurement. It is **declared**, and the measurement is recorded separately.

What is actually claimed

Precision about this matters, so here is the claim in full, with nothing left implicit.

The system asserts that tick zero is tick zero. That is a definition, and definitions are not the sort of thing that can be wrong.

It asserts, **separately and as metadata**, that tick zero is conventionally identified with the FLRW $t \to 0$ limit, and that the published uncertainty in that identification is ±0.020 Gyr, citing Planck 2018. This is a claim about the world and it may be revised.

And it asserts nothing whatever about whether time began, whether there was a first moment, or whether the FLRW limit corresponds to a physical event at all.

> **DEFERRAL IS NOT EVASION**
>
> You have now been told that the datum is a decision rather than a discovery, and why the short reason forces it. What Part IV adds is three further reasons the specification gives, one more that it does not, and — the part that turned this project into a book — what was **done** about it: how a claim can be declared, cited, given an exact magnitude, and simultaneously made impossible to compute with.
>
> If you are on the engineer's path and skipping Part IV, the sentence to carry forward is this one: the uncertainty in the age of the universe does not make the timestamps uncertain, because it is not an operand.

The datum is in ordinary company

One last thing, because "stipulated origin" can sound exotic and is not.

TAI's zero is 1 January 1958, chosen by agreement. The Julian Day epoch is 1 January 4713 BC, chosen because it made a convenient common multiple of three cycles. The Unix epoch is 1 January 1970, chosen because it was recent and round. None of these is a discovery about the universe; all of them are decisions that turned out to be useful.

> **INTERPRETATION**
>
> What is unusual here is not that the origin is stipulated. It is that the stipulation is **declared as such in the artifact**, with the physical claim held separately and marked non-computable — rather than being a fact about the system's history that you would have to go and read about.

> Most epochs are stipulated and silent about it. This one is stipulated and says so, in a place the compiler can reach.

WHAT THIS CHAPTER ESTABLISHED

- Tick zero is the datum: stipulated, conventionally identified with the FLRW $t \to 0$ limit, and explicitly not a measurement.
- The domain is unsigned. A request for a time before the datum is an error, not a negative number — a refusal of the question rather than an answer to it.
- Exactness cannot come from measurement: the ±0.020 Gyr uncertainty is a 58-digit number of ticks, and an exact count cannot inherit it.
- The physical claim is recorded separately as metadata, cited, and — as Part IV shows — made impossible to use in arithmetic.

CHAPTER 4

4

The universe second

A tick is far too small to think in. It is 5.4×10^{-44} seconds; the age of the universe is about 8×10^{60} of them. Nobody reads a 61-digit number.

So there is a ladder of larger units above the tick, and it has an unusual property: every rung is a whole power of five of the one below, and the whole ladder is a whole power of five of the tick. That property is what makes the notation work, and this chapter is about what it buys and what it costs.

TERM **Beat**

5^{60} ticks, about 46.762 milliseconds. The reference rung of the ladder, and what the specification calls the **universe second** — a unit of human-noticeable size with no Earth content whatever.

Why base five

Because $5^5 = 3125$, and 3125 is a number of exactly five base-5 digits.

That is the entire reason. It is worth being blunt about it, because a book that spends its second half among Pythagoreans and Neoplatonists has an obligation to say clearly where its numbers came from, and this one came from a digit-packing convenience.

RULE N

No constant, base, or magnitude in this system acquires significance by resembling a number in a tradition. Not the five, not the sixty in 5^{60}, not the 3125.

The research did, as it happens, go looking. The first place it looked — Losev on the Neoplatonic tetrad — had a five waiting, and the Pythagorean tetraktys sums to ten and builds from four. Those are interesting facts about those texts. They are not evidence about this software, they were not inputs to it, and the book reports them once, in Part VI, labelled as what they are.

The ladder

Each rung is $5^5 = 3125$ of the rung below, which is to say each rung is exactly five base-5 digits wide. Rung k is 5^{60+5k} ticks, indexed from the beat at $k = 0$.

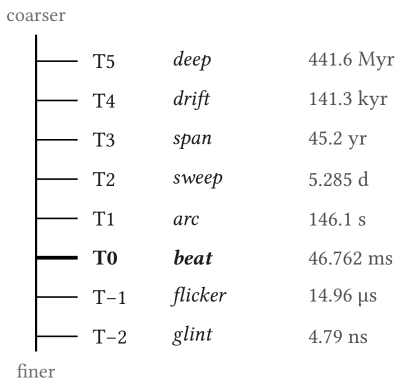

Figure 2 — The named rungs of the tier ladder. Each is 3125 of the one below. The ladder continues in both directions; only the named tiers are shown.

What the ladder buys

Three things, and they are all consequences of the same fact — that a timestamp is the tick count written in base 5 and grouped in fives.

Truncation is rounding. Write fewer groups and you have said the same thing less precisely. There is no separate rounding step, no scaling, no loss of a different kind: the digits you dropped are exactly the precision you gave up. This is the property that makes the uncertainty discipline of Part II possible at all.

Prefix comparison is chronological comparison. Two timestamps compare in the order their digits compare, because base-5 positional notation is monotone. Sorting the text sorts the times.

Writing all the digits pinpoints one tick. There is no accumulated conversion error between the coarse view and the fine one, because they are the same integer read at different widths.

Here is one real instant, at full precision and then at three coarser tiers:

ucal explain — one instant, four precisions

```
ticks
8070205189128471254993117657693008777530466139316558837890625

human     UC1 0031·0687·2481·3000·1638·3018:0779·2671·2006·1837·…
tiers:
  T5 deep   31
  T4 drift  687
  T3 span   2481
  T2 sweep  3000
  T1 arc    1638
  T0 beat   3018
```

The `0031` at the front is deeps since the datum. The `3018` before the colon is beats within the current arc. Every group is a base-5 number written in decimal, which is why none of them exceeds 3124.

INTERPRETATION

There is something worth noticing about that colon. It sits at the beat — at 5^{60} — and it is the only place in the notation where a human convenience has been allowed in. It is there because the beat is roughly the scale at which a person can notice a duration, so it is the natural place to put the decimal point of a time system built for beings who perceive at that scale.

That is a concession to the reader, not to Earth. It changes no arithmetic.

What the ladder costs

Nothing on this ladder is near anything you know.

A second is 21.385 beats — not a whole number, and not close to one. An hour is 24.6 arcs. A day is 0.189 sweeps. A year is 0.699 spans.

THE TWO SECONDS DO NOT DIVIDE

This is the sharpest form of the cost, and the system says so on its own ladder output:

> The two seconds are incommensurable above T−6. One bridge second is 21.385061835 beats, not a whole number, because `BEAT` carries 5^{60} while `SECOND` carries only 5^{30}. They share a common measure only at the tick.

So the beat is not a second in disguise, and no amount of rescaling would make it one. They agree at the tick and nowhere above it — which is exactly why the tick is primitive rather than either of them.

That is the honest consequence of leaving the Earth paradigm. If you want units with no planetary content, you do not get to keep the hour. The hour **is** planetary content.

The mark

The project's own mark is a diagram of this chapter, and it is worth reading as one.

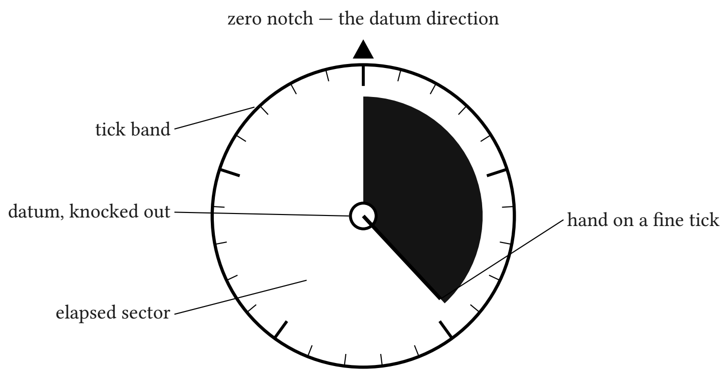

Figure 4 — The mark, read as a diagram. Five heavy divisions on the tick band, not twelve — the base is visible at a glance. The centre is negative space.

The centre dot has its core knocked out. That is deliberate: the datum is stipulated rather than observed, so the mark declines to draw anything there. The hand lands on a fine tick rather than a coarse division, because the claim the instrument makes is that it can pinpoint an arbitrary exact instant — not that it can round to a tidy one.

INTERPRETATION

A mark that showed a filled centre would be making a claim the specification refuses. The identity was drawn after the rule existed, and it obeys it. That is a small thing, and it is the sort of small thing this project is made of.

WHAT THIS CHAPTER ESTABLISHED

- The beat is 5^{60} ticks ≈ 46.762 ms — the **universe second**, a human-scale unit with no Earth content.
- Base five because $5^5 = 3125$ is five base-5 digits wide. Rule N: no further meaning, and the book says so before Part VI can be misread.
- The uniform ladder makes truncation **be** rounding, prefix comparison **be** chronological comparison, and full precision pinpoint one tick.
- The cost is that nothing on the ladder is near a second or an hour — one second is 21.385 beats, and the two share a measure only at the tick.

PART II

What was built

An engineer may stop at the end of this part and have received a complete technical article.

CHAPTER 5

5

The domain

NOT YET DRAFTED

This chapter is specified in RFC UCAL-A1 §16.2 and has not been written. Its outline is below so that the book's shape is visible and its length is not quietly understated.

Specified content

- Unsigned, closed, checked. Why 512 bits: 2.29×10^{103} years, past proton decay and heat death, with the present epoch using 7×10^{-17} of the range.

- Exact integer arithmetic and the refusal of floating point anywhere — what it costs (certified interval quadrature instead of a library call) and what it buys (a rigorous enclosure instead of a tolerance guess).
- Overflow as a typed error; no wrapping arithmetic exposed on time types.

CHAPTER 6

6

Notation

NOT YET DRAFTED

This chapter is specified in RFC UCAL-A1 §16.2 and has not been written. Its outline is below so that the book's shape is visible and its length is not quietly understated.

Specified content

- Two text forms — decimal tier groups for humans, base-5 digit groups as canonical — and why one value with two tagged forms is not redundancy.

- Canonical binary as fixed 64 bytes big-endian, so byte order is chronological order and the encoding is a usable database key.
- UCID as 52-character Crockford base-32 over 256 bits, with both caveats stated: undefined above 2^{256}, and carrying no randomness.

CHAPTER 7

7

The bridge

NOT YET DRAFTED

This chapter is specified in RFC UCAL-A1 §16.2 and has not been written. Its outline is below so that the book's shape is visible and its length is not quietly understated.

Specified content

- `SECOND` as the single declared bridge constant, an exact integer of ticks, so conversion into absolute time never rounds.

- TT as the only pivot; leap seconds confined to the parse/format boundary.
- Why an input finer than 10^{-30} s is rejected rather than rounded.
- The alignment invariants that fall out free: whole seconds land with thirty trailing base-5 zeros, whole nanoseconds with twenty-one.

CHAPTER 8

8

Derived calendars

NOT YET DRAFTED

This chapter is specified in RFC UCAL-A1 §16.2 and has not been written. Its outline is below so that the book's shape is visible and its length is not quietly understated.

Specified content

- Units derived from a body's rotation, solar day, and orbital period as exact rationals of ticks.

- Intercalation derived by continued-fraction expansion, never declared.
- The one empirical component — the anchor — flagged here, developed in chapter 16.
- The legacy/derived distinction, applied evenhandedly to the Gregorian and to scriptural chronologies alike.

PART III

What implementation refused

Design as something that loses arguments.

CHAPTER 9

9

Divergence

NOT YET DRAFTED

This chapter is specified in RFC UCAL-A1 §16.2 and has not been written. Its outline is below so that the book's shape is visible and its length is not quietly understated.

Specified content

- Which UCAL-1 rules survived contact with the compiler, which were altered, which were dropped, and what each change cost.

- Figure F10 — the divergence map, rules coloured by fate.

CHAPTER 10

10

The 97/400 correction

NOT YET DRAFTED

This chapter is specified in RFC UCAL-A1 §16.2 — the centrepiece and has not been written. Its outline is below so that the book's shape is visible and its length is not quietly understated.

Specified content

- Two RFC revisions asserted the machinery reproduces the Julian **and** Gregorian rules as convergents. The Julian rule is convergent 1.

- 97/400 is not a convergent at any depth. 8/33 is more accurate with a denominator twelve times smaller; 31/128 is 124× more accurate.
- The machinery contradicted its author, and the author published the contradiction — dated, including the revisions that were wrong.
- The direct answer to Maimonides' charge in **Guide** I.73 that such systems are built to make their conclusions provable.

PART IV

The datum

Why it cannot be measured, and what was done instead.

CHAPTER 11

11

Why it cannot be measured

NOT YET DRAFTED

This chapter is specified in RFC UCAL-A1 §16.2 and has not been written. Its outline is below so that the book's shape is visible and its length is not quietly understated.

Specified content

- Rule Q's three reasons: exactness cannot come from measurement; the FLRW $t \to 0$ limit is not an observable event; the extrapolation is model-dependent.

- Kant's fourth, which the specification does not state: the question presupposes a completed totality that is not a possible object of knowledge.
- The datum in ordinary company — TAI 1958, the Julian Day epoch, Unix — and the exact parallel to the SI second.

CHAPTER 12

12

What was done instead

NOT YET DRAFTED

This chapter is specified in RFC UCAL-A1 §16.2 and has not been written. Its outline is below so that the book's shape is visible and its length is not quietly understated.

Specified content

- `BIG_BANG_CLAIM` as a `SignedWindow` with no arithmetic implementations, `UCAL-E0025`, and the compile-fail test proving the type cannot reach an operand position — **the book's central exhibit**.
- `datum_provenance` as machine-readable, re-executable data, with its −0.017 190 364 s rounding residual.
- Kant's constitutive/regulative distinction: Kant policed the boundary with philosophical discipline; the crate polices it with a type.
- UC-Θ as unbuilt — the profile in which the datum is the beginning of time at organization, and what it would cost.

CHAPTER 13

13

Florensky's radius

NOT YET DRAFTED

This chapter is specified in RFC UCAL-A1 §16.2 and has not been written. Its outline is below so that the book's shape is visible and its length is not quietly understated.

Specified content

- **Мнимости в геометрии** (1922) as the closest precedent and the clearest cautionary tale in one text.

- The Dante geometry as genuine achievement; then §9, where a formal artifact is read as a physical place.
- The distance between the two is exactly one move — and Florensky had no rule against the second.
- Basil, **Hexaemeron** I.6, and the historical cost: persecution from 1922 to execution in 1937.

PART V

Any celestial body

How far the approach generalises, and precisely where it fails.

CHAPTER 14

14

Universal baselines

NOT YET DRAFTED

This chapter is specified in RFC UCAL-A1 §16.2 and has not been written. Its outline is below so that the book's shape is visible and its length is not quietly understated.

Specified content

- Nothing in the arithmetic references a rotation, an orbit, or a civil calendar.

- One derivation mechanism — every calendar is (Body, Anchor, Cycles, LeapRule) — with Earth as an ordinary instance and no crate named after a body.

CHAPTER 15

15

Deriving a calendar

NOT YET DRAFTED

This chapter is specified in RFC UCAL-A1 §16.2 and has not been written. Its outline is below so that the book's shape is visible and its length is not quietly understated.

Specified content

- Earth: convergents 1/4, 7/29, 8/33, 31/128 — the Julian rule as convergent 1, the Gregorian absent.

- Mars: 668.59 sols per year, convergent 16/27.
- Titan: tidally locked, handled with no special case — which is the point.
- The Metonic cycle 235/19 derived from Earth's periods unaided.

CHAPTER 16

16

Where it breaks

NOT YET DRAFTED

This chapter is specified in RFC UCAL-A1 §16.2 — no shorter than ch. 15 and has not been written. Its outline is below so that the book's shape is visible and its length is not quietly understated.

Specified content

- The anchor is empirical and cannot be derived.

- A body with no qualifying satellite has no month — Mars is the worked case, and the absence is the correct output.
- A rogue planet has no year.
- A tidally locked body's day equals its orbit.
- Relativistic environments are out of scope.
- Body parameters carry secular rates and validity windows and are wrong outside them — which matters most at exactly the deep-time scale this project targets.

CHAPTER 17

17

The one qualification

NOT YET DRAFTED

This chapter is specified in RFC UCAL-A1 §16.2 — one page and has not been written. Its outline is below so that the book's shape is visible and its length is not quietly understated.

Specified content

- For a single planet this is a curiosity. For timekeeping across two or more bodies, a universal ladder with local overlays may be the only coherent arrangement.

- A claim about coherence, not a recommendation for adoption.

PART VI

Nine readings

The traditions as readers of the artifact — neither validating it nor validated by it. Every chapter names its conflict.

CHAPTER 18

18

Greek

NOT YET DRAFTED

This chapter is specified in RFC UCAL-A1 §17.2 — A1 and has not been written. Its outline is below so that the book's shape is visible and its length is not quietly understated.

Specified content

- Archimedes' **Sand Reckoner** builds a positional hierarchy to make a cosmic magnitude expressible, arriving at $\sim 10^{63}$.

- Euclid X.2′s anthyphairesis **is** the intercalation algorithm.
- **Conflict:** Aristotle holds time a continuum and the instant a limit; the instrument is discrete. And **Physics** IV.14 makes the datum's objectivity depend on there being a counter.

CHAPTER 19

19

Jewish

NOT YET DRAFTED

This chapter is specified in RFC UCAL-A1 §17.2 — A2 and has not been written. Its outline is below so that the book's shape is visible and its length is not quietly understated.

Specified content

- **Molad tohu** — "the new moon of chaos" — a stipulated epoch placed before the event it anchors, nineteen centuries early.

- The **ḥelek** chosen for divisibility, exactly as `SECOND` was.
- **Conflict:** the compressed Persian period is provenance overruled by doctrine, and the crate's re-executable chain would expose it — so the instrument judges a tradition, which the book's own rules forbid the author from doing.

CHAPTER 20

20

Islamic

NOT YET DRAFTED

This chapter is specified in RFC UCAL-A1 §17.2 — A3 and has not been written. Its outline is below so that the book's shape is visible and its length is not quietly understated.

Specified content

- Ghazālī's **ʿāda** — divine custom yielding practical certainty without necessity — is what a validity window encodes.

- **Conflict:** under Premise 6 an orbital period is a habit, so a guaranteed drift bound over 400,000 years is meaningless. The crate's ontology is Rushdian and it took a side without arguing for it.

CHAPTER 21

21

Patristic and Latin

NOT YET DRAFTED

This chapter is specified in RFC UCAL-A1 §17.2 — A4 and has not been written. Its outline is below so that the book's shape is visible and its length is not quietly understated.

Specified content

- Basil's road-and-house image is Rule Q's content exactly.
- Augustine's answer is `UCAL-E0020`: an error, not a negative number.

- **Conflict: ex nihilo** is held firmly, so UC-Θ is heterodox by this standard — and the book must say so rather than soften it.

CHAPTER 22

22

Orthodox

NOT YET DRAFTED

This chapter is specified in RFC UCAL-A1 §17.2 — A5 and has not been written. Its outline is below so that the book's shape is visible and its length is not quietly understated.

Specified content

- Gregory of Nyssa's διάστημα makes a calendar an instrument of the created order by definition — the limitation is the doctrine, not a defect.

- Palamas' essence/energies distinction as the strongest licence the thesis receives from any tradition.
- **Conflict:** Rule N sides with the 1913 Synodal position against Florensky and Losev — the two thinkers this book most depends on.

CHAPTER 23

23

Latter-day Saint

NOT YET DRAFTED

This chapter is specified in RFC UCAL-A1 §17.2 — A6 and has not been written. Its outline is below so that the book's shape is visible and its length is not quietly understated.

Specified content

- D&C 130:4–5 states Rule K's thesis in 1843: reckoning is body-relative.
- Abraham 3:4 is a declared bridge constant with a profile tag on each side.

- **Conflict:** Kolob is a governing **body**, where Rule K.5 permits only an abstract ladder — the privilege is located differently.

CHAPTER 24

24

Modern philosophy

NOT YET DRAFTED

This chapter is specified in RFC UCAL-A1 §17.2 — A7 and has not been written. Its outline is below so that the book's shape is visible and its length is not quietly understated.

Specified content

- The First Antinomy supplies a fourth reason for Rule Q the specification does not state.

- Constitutive versus regulative, enforced by a type with no arithmetic: Kant policed the boundary by discipline, the crate by the compiler.
- **Conflict:** Kant's number is the schema of magnitude — monadic — which contradicts the eidetic number of A1 and A8, and Rule G asserts both readings of one integer.

CHAPTER 25

25

Russian

NOT YET DRAFTED

This chapter is specified in RFC UCAL-A1 §17.2 — A8 and has not been written. Its outline is below so that the book's shape is visible and its length is not quietly understated.

Specified content

- Fyodorov makes total addressability of the past an obligation.

- Vernadsky's process-relative times and Chizhevsky's solar-period dating as Rule K's unacknowledged ancestry, a century early.
- **Conflict:** Bugaev proposes discreteness **as worldview**, which the book declines — from the closest kin. And Losev's имя makes the name constitutive where Rule N makes it a locale-table entry.

CHAPTER 26

26

Method

NOT YET DRAFTED

This chapter is specified in RFC UCAL-A1 §17.2 — A9 and has not been written. Its outline is below so that the book's shape is visible and its length is not quietly understated.

Specified content

- Klein on **arithmos**; Sorabji on time, creation, and the continuum.

- **Conflict:** the method makes the code the invariant and the traditions the variables. That is itself a metaphysical choice, made without argument — and disclosing it is the last and most self-referential application of Rule M.

PART VII

The instrument as research tool

What a program can do that an argument cannot: it can be run against material its author did not choose, and return an answer he did not want.

CHAPTER 27

27

Six samples

NOT YET DRAFTED

This chapter is specified in RFC UCAL-A1 §18 and has not been written. Its outline is below so that the book's shape is visible and its length is not quietly understated.

Specified content

- S1 — comparative chronology on one axis: Seder Olam, Byzantine, Ussher, D&C 77, each a declared profile.

- S2 — calendar audit by convergent. Julian **is** convergent 1; Gregorian is not a convergent at any depth; Jalali 8/33 **is** convergent 3.
- S3 — uncertainty audit: cited versus stipulated.
- S4 — cross-body simultaneity: the demonstration that "now" is not a shared object.
- S5 — measuring διάστημα with no Earth content in the units.
- S6 — a revealed ratio evaluated as a bridge constant, refused as an anchor.

CHAPTER 28

28

Null results

NOT YET DRAFTED

This chapter is specified in RFC UCAL-A1 §16.2 and has not been written. Its outline is below so that the book's shape is visible and its length is not quietly understated.

Specified content

- Each sample's negative finding, stated as plainly as its positive one.

- Where a sample produced nothing, say so; where a result was weaker than hoped, say how much.
- This chapter is what distinguishes the book from a demonstration.

PART VIII

The claim

CHAPTER 29

29

Why a program

NOT YET DRAFTED

This chapter is specified in RFC UCAL-A1 §16.2 and has not been written. Its outline is below so that the book's shape is visible and its length is not quietly understated.

Specified content

- An essay can assert that a distinction ought to be respected; a type system can make violating it fail to build.

- Why prose cannot be **run** by someone who disagrees.

CHAPTER 30

30

Uselessness restated

NOT YET DRAFTED

This chapter is specified in RFC UCAL-A1 §16.2 and has not been written. Its outline is below so that the book's shape is visible and its length is not quietly understated.

Specified content

- The thesis, not an apology. Part V's qualification restated as a limit rather than a rescue.

CHAPTER 31

31

What is not claimed

> **NOT YET DRAFTED**
>
> This chapter is specified in RFC UCAL-A1 §16.2 and has not been written. Its outline is below so that the book's shape is visible and its length is not quietly understated.

Specified content

- The negative inventory, assembled from Appendix C.

CHAPTER 32

32

Kant's moon

NOT YET DRAFTED

This chapter is specified in RFC UCAL-A1 §16.2 and has not been written. Its outline is below so that the book's shape is visible and its length is not quietly understated.

Specified content

- Transcendental illusion is not error but a natural appearance that persists after diagnosis — the astronomer knows the moon is not larger at the horizon and still sees it that way.
- Every reader who sees a 61-digit integer will read it as a fact about being, and no rule prevents that.
- `UCAL-E0025` does not cure the illusion; it refuses to compute with it, which is the only thing a specification can do.

AFTERWORD

About the author

Vladimir Ulogov has spent decades building infrastructure for distributed systems — the kind of software that watches other software. Early in his career he worked on monitoring and telemetry platforms; later years took him into federated observability, telemetry buses, and the architecture of systems that have to make sense of millions of data points without losing the thread.

Observability, in the end, is a discipline of **coherence** — of never reporting a state the system cannot account for, of insisting that every signal follow from something real. It is not an accident that a calendar built to point at an origin it cannot measure carries the same instinct: it declares what it knows, declares what it has merely stipulated, and refuses to let the second quietly become the first.

What makes him slightly unusual in his corner of the industry is a tendency to write his own tools — not small utilities, but programming languages. The Bund language (its compiler, its VM, its document store, its parser) lives in a long series of Rust crates on crates.io. `rust_dynamic`, `rust_multistackvm`, `bundcore` — each is a building block that exists because the off-the-shelf options didn't fit the shape of the work. `ucal` grew the same way, from an irritation about units that no existing library was going to fix, because no existing library thought it was a problem.

A work of love

`ucal` is open source, under a permissive licence — you can read it, fork it, study it, modify it, and pass it on. It carries no analytics and no telemetry. It was not built to be sold and it was not built to be adopted; it was built because the question would not go away, and because the only way to find out whether a distinction can be enforced by a compiler is to try to enforce one.

A note on cooperation

Vladimir believes firmly in the human capacity for mutual help — that we make better work, and live better lives, when we share what we know and what we build. Open source is one of the most concrete expressions of cooperation our era has produced: code read, improved, and passed forward without payment, without permission, by people who will never meet.

This book is offered in the same spirit. If it helps you see one distinction more clearly — between what an instrument measures and what it merely points at — that is enough.

Where to find more

GitHub `@vulogov` — the source for `ucal`, Bund, and the dozen-plus Rust crates that carry the infrastructure. Issues and pull requests welcome.

LinkedIn `/in/vladimirulogov` — posts on observability, the occasional long-form essay.

YouTube `@vulogov` — talks and walkthroughs from the conference trail.

If you find a claim in this book that the source tree does not support, open an issue. That is the only kind of correction the book is built to receive, and it is the kind it most wants.

END OF THE BOOK